PROBLEM SET 4

Show **ALL WORK** to get full/partial credit. Begin each problem on a new page, and clearly label each part of the problem. (**Max Score:** 50)

1) **(50 pts)** Consider the following operators on a Hilbert space $\mathbb{V}^3(C)$

$$L_x = \frac{1}{\sqrt{2}}\begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}, \quad L_y = \frac{1}{\sqrt{2}}\begin{bmatrix} 0 & -i & 0 \\ i & 0 & -i \\ 0 & i & 0 \end{bmatrix}, \quad L_z = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{bmatrix}$$

(a) **(5 pts)** What are the possible eigenvalues one can obtain if L_z is measured?

(b) **(15 pts)** Take the state in which $L_z = 1$. In this state what are

(i) **(5 pts)** $\langle L_x \rangle$

(ii) **(5 pts)** $\langle L_x^2 \rangle$

(iii) **(5 pts)** ΔL_x

(c) **(5 pts)** Find the normalized eigenstates and eigenvalues of L_x in the L_z basis.

(d) **(5 pts)** If the particle is in the state $|L_z = -1\rangle$, and L_x is measured, what are the possible outcomes, i.e. the probabilities $P(L_x = -1)$, $P(L_x = 0)$ and $P(L_x = 1)$?

(e) **(15 pts)** Consider the state $|\psi\rangle = \begin{bmatrix} 1/2 \\ 1/2 \\ 1/\sqrt{2} \end{bmatrix}$ in the L_z basis. If L_z^2 is measured in this state and a result +1 is obtained .

(i) **(5 pts)** what is the state after the measurement ?

(ii) **(5 pts)** how probable was this result ?

(iii) **(5 pts)** if L_z is measured, what are the outcomes and respective probabilities ?

(f) **(5 pts)** A particle is in a state for which the probabilities are $P(L_z = 1) = 1/4$, $P(L_z = 0) = 1/2$, and $P(L_z = -1) = 1/4$.

Show that the most general, normalized state with this property is

$$|\psi\rangle = \frac{e^{i\delta_1}}{2}|L_z = 1\rangle + \frac{e^{i\delta_2}}{\sqrt{2}}|L_z = 0\rangle + \frac{e^{i\delta_3}}{2}|L_z = -1\rangle$$

Hint: the argument can be made that adding an overall complex phase to the state does not affect its relative probabilities. For example, read text on p. 118 of Shankar referring to this for making the argument. Show that the relative probabilities are unaffected.